

LACKLAND AFB, TX, USA

WMO: 722535

Lat: 29.383N

Lon: 98.583W

Elev: 210

StdP: 98.82

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12909

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	0.5	-10.8	1.5	5.9	-8.3	1.9	7.6	11.4	13.3	9.9	12.3	3.0	340	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.8	38.0	23.4	37.3	23.5	36.2	23.5	26.6	31.2	26.0	30.7	25.5	30.4	4.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	21.4	27.8	25.0	20.6	27.5	24.3	19.7	27.1	84.3	31.3	81.8	30.4	79.4	30.5	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.5	7.7	DB	-5.9	40.1	3.8	1.4	-8.7	41.2	-10.9	42.0	-13.0	42.8	-15.8	43.9	
			WB	-6.8	27.1	3.4	0.8	-9.2	27.7	-11.2	28.2	-13.2	28.7	-15.6	29.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.4	11.7	14.3	17.3	21.4	25.6	28.5	29.6	29.9	27.0	22.0	16.4	12.4
	DBStd	7.63	5.01	5.45	4.80	4.00	3.19	2.23	1.83	1.73	2.89	4.28	4.95	4.96
	HDD10.0	106	40	21	6	1	0	0	0	0	0	1	8	30
	HDD18.3	743	212	133	76	17	1	0	0	0	1	17	92	194
	CDD10.0	4260	92	139	234	341	483	556	609	617	509	373	201	104
	CDD18.3	1857	6	19	46	108	226	306	351	359	259	132	35	10
	CDH23.3	20479	43	137	303	905	2209	3589	4389	4575	2843	1199	231	54
Wind	WSAvg	3.6	3.3	3.7	4.0	4.2	4.3	4.0	3.7	3.4	3.1	3.4	3.2	3.2
	PrecAvg	772	48	44	58	66	99	84	67	50	79	78	49	48
	PrecMax	1264	132	167	180	182	230	250	306	203	227	276	222	336
	PrecMin	315	1	3	0	7	6	6	2	0	6	0	0	4
	PrecStd	266	38	40	37	48	65	63	82	58	56	79	56	65
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.9	30.1	31.1	34.9	37.1	39.0	39.3	39.2	37.8	34.0	29.2	27.1
		MCWB	18.2	17.3	18.7	20.7	23.3	23.9	23.2	23.4	22.7	22.4	21.2	19.3
	2%	DB	24.3	27.0	28.4	32.1	35.0	37.3	37.9	38.0	36.0	32.2	27.6	25.0
		MCWB	16.5	17.6	19.1	21.7	23.5	23.9	23.2	23.6	23.0	22.6	20.0	18.2
	5%	DB	22.5	24.8	26.6	30.0	33.3	35.9	37.0	37.2	34.6	30.5	26.1	22.8
		MCWB	16.4	17.4	19.2	20.7	23.4	23.7	23.3	23.6	22.9	21.8	19.3	18.0
	10%	DB	20.5	22.6	24.8	28.0	31.7	34.1	35.6	36.0	32.9	29.0	24.2	21.2
		MCWB	15.5	16.9	18.3	20.3	23.1	23.8	23.6	23.7	23.0	21.3	19.2	17.0
	0.4%	WB	22.5	21.7	23.3	26.0	26.7	27.3	26.9	27.2	26.8	25.7	23.6	22.4
		MCDB	23.9	24.5	26.3	29.8	29.8	31.1	32.2	34.2	30.3	28.8	25.7	24.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	19.9	20.6	21.8	24.3	25.7	26.5	25.8	26.3	25.7	24.8	22.6	21.0
		MCDB	22.1	23.0	25.1	28.3	30.4	31.1	31.0	31.2	29.7	28.1	25.2	23.2
	5%	WB	18.6	19.8	20.8	23.0	25.0	25.8	25.5	25.6	25.2	24.1	21.7	19.8
		MCDB	20.8	22.5	23.8	27.0	30.2	30.8	30.7	30.5	29.4	27.5	24.2	21.8
	10%	WB	17.0	18.8	20.1	22.3	24.3	25.3	25.1	25.2	24.7	23.4	20.6	18.0
		MCDB	19.2	21.4	23.1	26.0	29.1	30.5	30.3	30.1	29.1	26.8	23.3	20.0
Mean Daily Temperature Range	5% DB	MDBR	12.5	12.2	11.8	11.9	10.9	10.9	11.5	11.8	11.2	11.7	11.7	12.1
		MCDBR	15.4	14.6	13.1	13.7	12.4	12.8	13.5	13.3	12.8	11.9	12.4	14.1
		MCWBR	9.4	8.2	6.6	6.0	3.8	2.9	2.9	2.9	3.3	4.6	6.7	8.6
	5% WB	MCDBR	11.2	10.6	10.1	10.2	10.4	10.4	11.1	11.1	9.9	8.9	9.6	10.4
		MCWBR	8.3	7.4	6.3	5.2	4.0	3.4	3.0	3.1	3.2	4.3	6.3	8.1
Clear-Sky Solar Irradiance	taub		0.327	0.334	0.358	0.400	0.446	0.429	0.435	0.421	0.414	0.374	0.360	0.319
	taud		2.499	2.482	2.431	2.327	2.245	2.348	2.362	2.385	2.355	2.457	2.430	2.532
	Ebn at Noon		912	934	930	895	851	860	853	865	861	879	864	904
	Edh at Noon		90	100	112	129	140	126	124	120	120	102	96	84
All-Sky Solar Radiation	RadAvg		3.03	3.56	4.50	5.34	5.76	6.61	6.65	6.37	5.22	4.33	3.25	2.69
	RadStd		0.34	0.64	0.47	0.54	0.60	0.43	0.58	0.36	0.52	0.57	0.37	0.35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+88

Nomenclature: See separate page